



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION

NAAC
GRADE A
Accredited University

Minor Project

1010103494

Contextual Awareness Recommendation and Transaction System

Group :

Akshat Shah - 2201031000159

Yash Dalsaniya - 2201031000160

Meet Sharma - 2201031000175

Branch : Information Technology

Guide Name : Prof. Purvi Patel

Introduction

Navigating the Deluge of Choices

In today's retail and e-commerce landscape, an overwhelming volume of transaction data is generated daily. While this data offers immense potential, it also presents significant challenges for both consumers and businesses. Customers frequently experience "choice overload," making it difficult to select relevant products from a vast inventory. Simultaneously, businesses often rely on traditional recommendation systems that, while useful, are typically limited to basic metrics like purchase frequency or collaborative filtering.

CARTS aims to revolutionize this approach by providing smarter, context-aware recommendations that integrate not just purchasing patterns, but also crucial profitability measures.

Background and Motivation

Beyond "People Also Bought"

Traditional recommendation systems, exemplified by Amazon's "people also bought" feature, predominantly rely on historical co-purchase data. While effective in identifying popular product associations, these systems possess significant limitations:

- **Profit Blindness:** They do not consider the inherent profit margins of individual items.
- **Equal Treatment:** All products are treated as equally valuable, irrespective of their actual business contribution.
- **Unbalanced Objectives:** They often fall short in balancing customer satisfaction with critical business objectives like profit maximization.

Motivation: Retailers urgently need a system that can both maximize profits and maintain recommendation relevance. By incorporating detailed metrics such as cost price, selling price, profit percentage, and margin, CARTS can move beyond mere frequency to suggest high-value products that drive significant business growth.

Relevance and Importance

Broad Relevance

CARTS is invaluable for a wide array of businesses, from supermarkets and large retail chains to dynamic e-commerce platforms. It empowers them to implement superior cross-selling strategies, directly leading to higher revenue generation.

Strategic Importance

CARTS introduces a crucial new dimension to recommendation systems: [profitability awareness](#). This capability not only helps optimize inventory by focusing on high-turnover, high-margin items but also ensures customers receive truly personalized and meaningful suggestions.

Competitive Edge

In a crowded marketplace, the ability to recommend products that are both desired by the customer and profitable for the business provides a significant competitive advantage. CARTS helps businesses achieve this delicate balance.

Bridging the Research Gap

Literature Review

Existing Approaches:

- **Collaborative Filtering:** Relies on user-item similarity but overlooks item-level business attributes like profit.
- **Content-Based Filtering:** Focuses on product features but struggles to capture complex transaction-level patterns.
- **Apriori Algorithm:** Effective in finding frequent itemsets, but traditionally applied without consideration for business profitability metrics.

Research Gap:

Most current systems heavily prioritize **frequency** and **support** metrics, often completely overlooking the critical aspect of an item's **profit contribution**.

CARTS' Contribution:

CARTS significantly extends the well-established Apriori algorithm by ingeniously integrating **profit margins, cost, and custom bias weights** directly into the recommendation generation process. This ensures that recommendations are not just popular, but also strategically valuable.

Core Objectives of CARTS

01

Develop a Balanced Recommendation System

To create a robust system that intelligently balances customer purchasing patterns with critical business profitability goals.

02

Compute Item Weights & Biases

To accurately calculate and assign numerical weights and biases to each product, incorporating its Cost Price (CP), Selling Price (SP), Profit Percentage, and Margin.

03

Apply Enhanced Apriori Algorithm

To leverage the Apriori Algorithm for identifying strong association rules, seamlessly integrating the newly computed profit-based weights.

04

Generate Smart Recommendations

Recommended items are not only frequent and relevant to customer behavior but also strategically valuable and highly profitable for the business.

05

Ensure Scalability

To design a flexible and adaptable framework capable of scaling to large datasets and diverse real-world retail applications.

Tools & Technologies

Powering CARTS



- **Programming Language:** Python – Chosen for its extensive libraries and ease of use in data science.
- **Key Libraries & Packages:**
 - **Pandas & NumPy:** Essential for efficient data preprocessing, manipulation, and numerical calculations.
 - **mlxtend:** Specifically for implementing the Apriori algorithm and extracting association rules.
 - **Matplotlib:** For creating clear and insightful data visualizations to understand patterns and results.
- **Dataset Requirements:**
 - **Item Dataset:** Must include critical attributes like Cost Price (CP), Selling Price (SP), Profit Percentage, and Margin for each product.
 - **Transaction Dataset:** A substantial collection of at least 1000 transactions, with each containing 2-5 items to ensure sufficient data for pattern recognition.
- **Development Environment:** Google Colab

Method and Approach

1. Data Preparation

Clean and standardize item and transaction datasets, ensuring consistency in CP, SP, Profit %, and Margin.

2. Weight & Bias Computation

Assign numerical importance to each item, balancing its frequency with its profitability contribution.

3. Apriori Algorithm

Apply frequent itemset mining and extract association rules based on support, confidence, and lift.

4. Contextual Awareness

Re-rank identified rules by integrating profit-based weights, filtering for those that boost both sales and profit.

5. Recommendation Generation

Output tailored item combinations and recommendations for enhanced customer experience and retailer benefit.

Transformative Outcomes

Expected Impact of CARTS



Profit-Aware Recommendations

Move beyond simple frequency to deliver a sophisticated recommendation system that directly factors in product profitability.



Boosted Cross-Selling

Significantly enhance cross-selling potential for retailers by suggesting complementary items that also yield higher margins.



Balanced Approach

Achieve a harmonious balance where customers receive genuinely meaningful suggestions, and businesses concurrently maximize their profitability.



Scalable Model

Provide a robust and scalable framework that can be seamlessly extended to accommodate larger datasets and diverse industries, including e-commerce, Fast-Moving Consumer Goods (FMCG), and large retail chains.

Further Reading & Research

Bibliography

- Researched Datasets from Kaggle:<https://www.kaggle.com/datasets/irfanasrullah/groceries>
- Viewed Documents on Apriori Algorithm and Recommendation Systems from:
 - <https://www.geeksforgeeks.org/machine-learning/apriori-algorithm/>
 - <https://www.geeksforgeeks.org/machine-learning/what-are-recommender-systems/>

